

Eigenvalues and inverse problems

Symmetric minimizer for a positive definite Jordan–Kronecker product

Difficulty: challenging **Importance:** interesting to the community**Status:** Solution claimed**Topic:** Structured eigenvalue problems and semidefinite optimization.**Rating rationale:** Challenging reflects a structured minimum-eigenvalue comparison for arbitrary positive definite pairs; community impact connects Kronecker eigenproblems and semidefinite optimization.**Resolution claim — 2026-09-11****Affirmative resolution claimed; submitted by Matthew Colbrook.** See the [complete manuscript](#), **Theorem SP-05, sections 1–3** ([PDF](#) · [LaTeX](#)), prepared 11 September 2026.

For arbitrary real symmetric positive definite A, B , the manuscript claims a nonzero real positive-semidefinite eigenmatrix for the smallest eigenvalue of $X \mapsto AXB + BXA$. Section 3 derives the exact symmetric/skew-symmetric Rayleigh-quotient inequality below. No commutativity, rank restriction or simple-eigenvalue assumption is imposed.

The manuscript discloses generation in a ChatGPT conversation. Its full proof has not been independently verified here, and no publication or priority claim is made. [Supporting diagnostics](#) were rerun successfully; these are not an independent proof audit. This update checks the submitted claim against the displayed target; the dated literature checks below remain the record of the earlier search. The ratings above are historical, and this entry is excluded from the open count.

Problem statement

Let $n \geq 2$ and let $A, B \in \mathbb{R}^{n \times n}$ be symmetric positive definite. Let $T \in \mathbb{R}^{n^2 \times n^2}$ be the commutation matrix defined by $T \operatorname{vec}(X) = \operatorname{vec}(X^T)$, where vec stacks columns. Is it always true that

$$\min_{\substack{u \in \mathbb{R}^{n^2} \setminus \{0\} \\ Tu = u}} \frac{u^T (A \otimes B) u}{u^T u} \leq \min_{\substack{w \in \mathbb{R}^{n^2} \setminus \{0\} \\ Tw = -w}} \frac{w^T (A \otimes B) w}{w^T w}?$$

Equivalently, must the smallest eigenvalue of $A \otimes B + B \otimes A$ have an eigenvector $\operatorname{vec}(X)$ with $X = X^T \neq 0$?

Why it matters

An affirmative answer would identify which invariant subspace contains the smallest eigenvalue of an important structured matrix operator, relevant to positivity checks in semidefinite optimization.

References

- N. Kalantarova and L. Tunçel, [On the spectral structure of Jordan–Kronecker products of symmetric and skew-symmetric matrices](#), *Linear Algebra and its Applications* 608 (2021), 343–362. Conjecture 1, equation (7), p. 12 of arXiv v3; the preceding discussion explains the optimization connection.
- D. Kressner and B. Vandereycken, [A counterexample to the symmetric-maximizer conjecture for Lyapunov operators](#), 2026, abstract. This resolves a different symmetric-maximizer conjecture and is included to distinguish the recent result from this problem.

Earlier status check — 2026-09-08

On 2026-09-08, checked arXiv v3 (2020-08-08, still the latest listed version) and searched combinations of “Jordan-Kronecker”, “positive definite”, “conjecture”, “smallest”, “counterexample”, and the authors’ names, including 2025–2026 results. No resolution of Conjecture 1 was located. The source’s counterexamples to general symmetric-matrix interlacing do not dispose of its separately stated positive definite conjecture. The August 2026 Lyapunov result concerns an operator norm maximizer, rather than this positive definite Jordan–Kronecker minimum. No recent explicit reaffirmation of this exact conjecture was found; the status evidence is therefore bounded.

Audit update — 2026-09-10

Rechecked the [author manuscript](#), Conjecture 1 and equation (7). Its positive definite, minimum-eigenvalue formulation matches this entry. Searches for the Jordan–Kronecker symmetric-minimizer conjecture and later counterexamples found no resolution. Results for indefinite inputs or a maximum-eigenvalue objective do not settle the displayed claim.